Learning to Segment Referred Objects from Narrated Egocentric Videos

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Figure placeholder: NVOS overview with egocentric frames and referred-object masks (as in Fig. 1)

Problem & Contributions

- They introduce NVOS: given an egocentric clip and narration, segment per-frame masks for each referred noun phrase without spatial labels.
- They propose ROSA, a weakly-supervised pixel-level grounding framework aligning phrases with SAM mask proposals using CLIP-based dual encoders.
- They present VISOR-NVOS, a benchmark with 14,612 clips and object-linked narrations built on VISOR masks, and demonstrate zero-shot performance on VISOR-NVOS and VOST with transfer to YouCook2.

Related Work

Video Object Grounding

- Mostly third-person; typically outputs boxes or points.
- Egocentric grounding prior work predicts rectangular regions.

Video Narration Grounding

- Segments nouns in third-person videos with mouse traces.
- Their method targets egocentric videos without spatial labels.

VOS & Open-Vocabulary Segmentation

- Semi-supervised VOS relies on first-frame masks; no language grounding.
- Open-vocabulary segmentation predicts category masks, not instance disambiguation.

Phrase Grounding & CLIP

- Prior works often output boxes/points; adapting CLIP to regions/phrases is crucial.

ROSA: Method Overview

Video Frames $I_1..I\_T$ → SAM Mask Proposals $P_{t,m}$ + confidences → Context-Aware Region Encoder (Mask-Guided CLIP ViT + MLP) ⇒ $r_{t,m}$

Narration S with noun phrases $o_1..o\_N$ → Context-Aware Phrase Encoder (phrase-only EOT ⊕ narration-aware pooling) ⇒ w_n → Similarity g(r,w) × mask conf.

Local: Region–Phrase Alignment (R2P InfoNCE + P2R MIL-NCE with aligned regions) → Global: Video–Narration Alignment via Temporal Adaptive Pooling ⇒ ϕ(V,S)

Inference: per phrase n, per frame t, select $\mathrm{argmax}_m\ g(r_{t,m}, w_n)$

Backbone: CLIP ViT (image/text encoders frozen) Train loss: $L = L_{VNA} + \lambda L_{RPA}$

Encoders: Region and Phrase

Frame I_t → CLIP ViT L-1 layers (shared) → Last ViT layer with attention mask ⇒ CLS_P

Mask $P_{t,m}$ → MLP projection ⇒ $r_{t,m}$

Narration S → Text encoder + token pooling ⇒ S-aware

Phrase o_n → Text encoder on o_n ⇒ EOT (localized)

→ Average(S-aware, localized) ⇒ w_n

Method

Encoder Schematics

Training Details

- Data: Ego4D subset with 250k cooking-related clips; 4 frames sampled per 1s clip; one narration sampled per clip per iteration.
- Masks: SAM with 32×32 point grid; NMS 0.9; CLIP ViT-L/14 (best) and ViT-B/16 (ablations); λ = 0.5; 3-layer MLP for region encoder; CLIP encoders frozen.
- Losses: Global InfoNCE on video–narration affinity; Local R2P InfoNCE + P2R MIL-NCE on pseudo-aligned regions.

Temporal Adaptive Pooling

Frame-phrase scores ψ(I_t,o_n) → Weights softmax(ψ/α) → Weighted sum ⇒ σ(V,o_n)

Chart placeholder: α schedule $\alpha = \alpha_0$ β^{-s} (starts low → increases), shifting from single best frame to multi-frame aggregation

Training

Optimization

Datasets & Evaluation Setup

VISOR-NVOS

- 14,612 video clips; 37,170 referred objects.
- Narrations link noun phrases to VISOR object IDs.
- Avg 12.79 words; 2.54 objects/narration; max 9; splits: 7,561 val / 7,051 test.

VOST

- Egocentric with dramatic object transformations; one class per video.
- Metrics: J union and J ins due to instance ambiguity.

- Zero-shot evaluation on VISOR-NVOS (J, F, J&F) and VOST (J union, J ins) after training on Ego4D.

Datasets

VISOR-NVOS • VOST

Results Summary

- On VISOR-NVOS, their method (ROSA, ViT-L/14) achieves J=38.7, F=46.0, J&F=42.4, outperforming SAM+CLIP and CoMMa+SAM, and surpassing open-vocabulary baselines (ODISE, GroundedSAM) trained with stronger spatial supervision.
- On VOST (zero-shot), their method attains J union=23.2 and J ins=26.7, improving over SAM+CLIP and CoMMa+SAM and showing robustness to transformations.
- The best SAM proposal upper bound remains higher, indicating headroom and task difficulty.

VISOR-NVOS J: 38.7 | VISOR-NVOS F: 46.0 | VISOR-NVOS J&F: 42.4 | VOST J union: 23.2 | VOST J ins: 26.7

Results

Key Metrics

Limitations

- Frame-wise inference without explicit temporal consistency; integrating temporal context could improve performance.
- Training and evaluation focus on cooking; cross-domain gap persists despite improvements on non-cooking domains.

References, Acknowledgements, and Takeaway

- Key datasets/models: Ego4D, VISOR, VOST, SAM, CLIP; benchmark: VISOR-NVOS.
- Related methods: GroundedSAM, ODISE, CoMMa, MIL-NCE, ViT.
- Acknowledgements: T. Nagarajan, T. Afouras, Y. Song, A. Miller, J. Hu.

Single-Sentence Takeaway: Weakly-supervised region–phrase alignment with CLIP-based encoders and global–local contrastive learning enables zero-shot, pixel-level grounding in egocentric videos that surpasses strong baselines without using any spatial annotations.
Impact: Reduces annotation costs for fine-grained video understanding and provides a scalable path toward instance-level grounding across domains.